

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Marine Technology

TMR4240 Marine Control Systems I

Project Part 2
Design of a Dynamic Positioning System

Python-Based Simulation and Control Design Guide

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1 General Information

The report for Project Part 2 must be submitted on Canvas by the deadline stated in the Lecture Plan. The submission must include the project report and either a ZIP archive or a repository link containing the executable Python code used for all simulations performed by the group. The appendix must briefly describe how the simulations can be executed and how the reported results can be reproduced.

Project Part 2 will be evaluated using the same general principles as Project Part 1. The assessment therefore considers not only whether the implementation works, but also the quality of the technical work, the clarity and correctness of the explanations, the quality of the presented results, and the ability to justify engineering decisions. Particular attention should be given to how simulation results are plotted, interpreted, and discussed. The project is done in groups of three or four students, and only one report should be submitted per group. The Part 2 report is a self-contained document. It is read and graded on its own. The recommended length is about 10-20 pages excluding appendices. Use the IEEE two-column format.

Project Objective

The main objective of Project Part 2 is to extend the basic dynamic positioning (DP) system developed in Part 1 with detailed environmental loads, constrained thrusters, and observer-based feedback, and to validate the resulting complete system through reproducible simulations.

The project should be approached as an engineering development and validation process rather than simply as an implementation task. Students are expected to develop an understanding of how the different components of a DP system interact and to be able to explain and justify the modelling, control, simulation, and tuning choices made throughout the project.

A complete DP simulation involves several interconnected components. Consequently, the development should follow a structured process in which individual components are implemented and tested before the complete system is integrated.

Recommended Engineering Methodology

The project should generally follow the development process

Model → Implement → Test → Tune → Validate → Integrate → Analyse

Each subsystem should first be considered independently before being integrated into the complete DP system. Students should:

1. understand the purpose and assumptions of each subsystem;
2. implement the required functionality;
3. perform appropriate tests to verify that the implementation behaves as expected;
4. tune relevant parameters and explain the basis for the selected values;
5. validate the subsystem using suitable simulation cases;

6. integrate the validated subsystems into the complete DP system;
7. evaluate the behaviour of the complete system under relevant operating conditions;
and
8. analyse and discuss the obtained results, including limitations and possible improvements.

The report should demonstrate this development process rather than only presenting the final implementation. Important modelling assumptions, design decisions, parameter choices, tuning procedures, and validation tests should be clearly documented and technically justified.

Simulation Results and Technical Discussion

Simulation results must be presented in a clear and technically meaningful manner. Figures should be appropriately labelled, contain relevant units and legends, and be sufficiently detailed to support the discussion.

Results should not be presented without interpretation. For each important simulation, the report should explain what was tested, why the test was performed, what behaviour was expected, and what can be concluded from the obtained results. Where appropriate, quantitative performance measures should be used to support the conclusions.

The discussion should also identify limitations and conditions under which the implemented system may not perform as intended. A successful result should therefore not only be shown, but also explained in terms of the underlying system behaviour.

Reproducibility

All simulations used to produce results presented in the report must be reproducible from the submitted code. The code should therefore be organized such that another student or member of the teaching staff can understand how the simulations are configured and executed.

The appendix should provide the necessary instructions for running the simulations and reproducing the main results. Any required dependencies, configuration steps, or relevant assumptions should be stated clearly.

Teamwork and Individual Responsibility

This is a teamwork project, but each student is individually responsible for contributing actively and professionally to the project. The division of responsibilities agreed for Part 1 should be reviewed for the Part 2 tasks (wind extensions, observers, constrained allocation, simulations) and any change communicated to the teaching staff.

Each student is expected to take ownership of their assigned tasks, maintain communication with the group, and seek assistance when difficulties arise. The division of responsibilities should support parallel development where appropriate, while ensuring that all group members understand how their work contributes to the complete DP system.

Although responsibilities may be divided between group members, the final system is a common group deliverable. Students should therefore communicate interface requirements, implementation changes, and relevant technical findings to the rest of the group throughout the development process.

Handling Difficulties and Incomplete Results

Unforeseen problems, unsuccessful approaches, and incomplete implementations are a normal part of engineering development. Such issues should be identified, documented, and communicated at an early stage.

If a component or part of the project cannot be completed as intended, this should be clearly described in the report. The report should explain what was attempted, what difficulties were encountered, when the problem was identified, and what was learned from the attempt. Where possible, students should also discuss potential causes of the problem and suggest how the implementation or methodology could be improved.

A technically well-documented unsuccessful attempt, supported by appropriate analysis and reflection, is preferable to an unexplained result or an implementation whose behaviour cannot be justified.

Individual Contributions and Assessment

The final report must include an individual contribution statement describing the responsibilities and contributions of each group member. The assessment may take individual contributions into account.

Individual assessment will consider not only the final group result, but also evidence of active participation, technical understanding, and professional engagement with the project. Students are expected to contribute consistently, communicate relevant difficulties in a timely manner, and take responsibility for their assigned tasks.

The overall objective is therefore not simply to obtain a working simulation, but to demonstrate an engineering approach in which the system is developed systematically, tested and validated, and supported by clear technical reasoning and reproducible results.

Final Note

This project is an opportunity not only to apply the concepts introduced in the lectures, but also to develop the engineering skills needed to approach complex problems systematically. You will likely encounter challenges along the way, but working through these challenges is an important part of the learning process.

Take the time to understand what you are doing, ask questions when something is unclear, learn from unsuccessful attempts, and support each other throughout the project. By the end of the course, the goal is not only to have a working DP system, but to have gained a deeper understanding of marine control systems and greater confidence in solving engineering problems independently.

Good luck, work well together, and enjoy the learning process!

- Teaching staff

2 System Overview

A survey of DP architectures and control approaches is given in [3].

Project Part 2 builds directly on Project Part 1. The same vessel, coordinate frames, process plant, and basic DP structure are retained. Part 2 extends the system with more detailed environmental loads, observer-based state feedback, and realistic thrust and azimuth limitations. The supplied Python project is available from:

NTNU-MCS/TMR4240-LAB

The overall system architecture is shown in Fig. 1. The simulation consists of the vessel model, environmental disturbance models, measurement and observer blocks, control modules, thrust allocation, and actuator dynamics. The work comprises:

- the current model from Part 1, transferred without changing its physical conventions;
- the Part 1 wind model extended with gust speed and slowly varying direction;
- the Part 1 DP controller and reference model, retuned or extended only when justified;
- the Part 1 thrust allocation extended for the Part 2 constraints;
- two different observer designs (new); and
- the supplied irregular-wave, measurement, actuator, and vessel models.

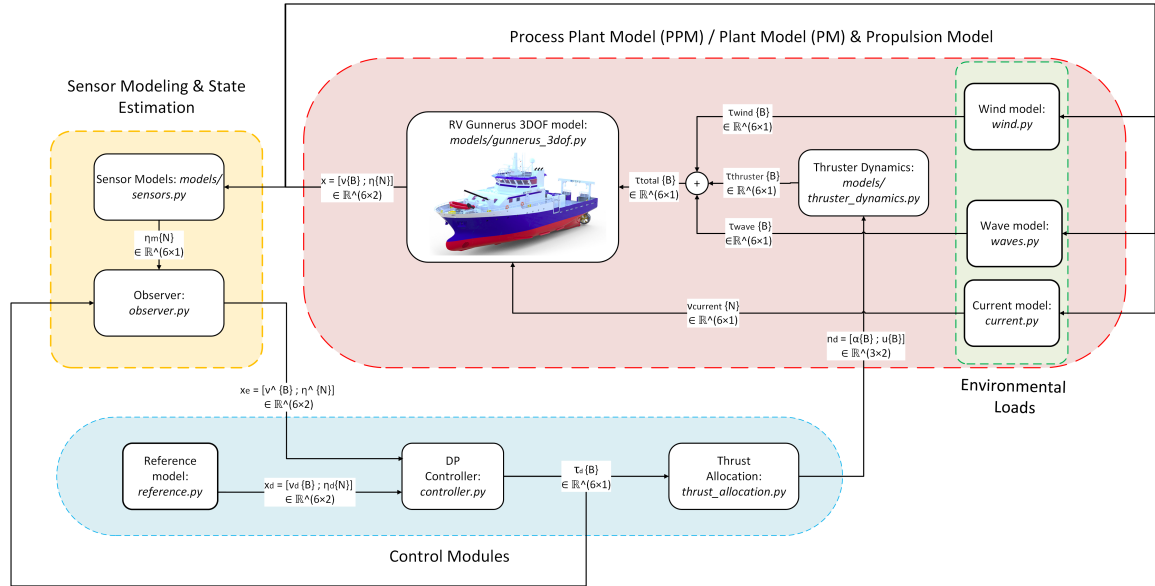


Figure 1: Conceptual overview of the DP-controlled Gunnerus vessel, showing the control modules, thruster dynamics, vessel dynamics, environmental disturbances, and feedback of the vessel state between the NED frame $\{N\}$ and body-fixed frame $\{B\}$.

The supplied simulator handles vessel-state integration, the measurement layer, actuator dynamics, and logging. The repository documentation gives the current software layout and execution instructions; this assignment defines the stable modelling, design, simulation, and report requirements.

3 Coordinate Frames, Notation, and Modelling Assumptions

Use this section of your report to define notation and assumptions before presenting any design. A reader should be able to understand the complete report without opening any simulation files. All symbol definitions must be consistent across sections. *This section was solved in Project Part 1; copy it into the Part 2 report where it still holds. State which of them Part 2 changes; in particular, thruster dynamics and their limits are enabled in Part 2.*

3.1 Coordinate Frames

Two coordinate frames are used to describe the vessel motion: an Earth-fixed navigation frame and a body-fixed vessel frame.

The **Earth-fixed NED frame** is denoted by n and is fixed to the Earth. Its axes are defined as x_n pointing North, y_n pointing East, and z_n pointing Down. This frame is used to describe the vessel position and heading relative to the surrounding environment.

The **body-fixed frame** is denoted by b and is attached to the vessel. Its origin is typically chosen at the vessel centre of gravity or at another clearly specified reference point. The x_b axis points forward toward the bow and corresponds to surge motion, the y_b axis points to starboard and corresponds to sway motion, and the z_b axis points downward.

Both coordinate frames are right-handed. With the NED convention, a positive rotation about the downward z axis is clockwise when the vessel is viewed from above. Consequently, the yaw angle ψ is measured clockwise from North to the vessel bow direction.

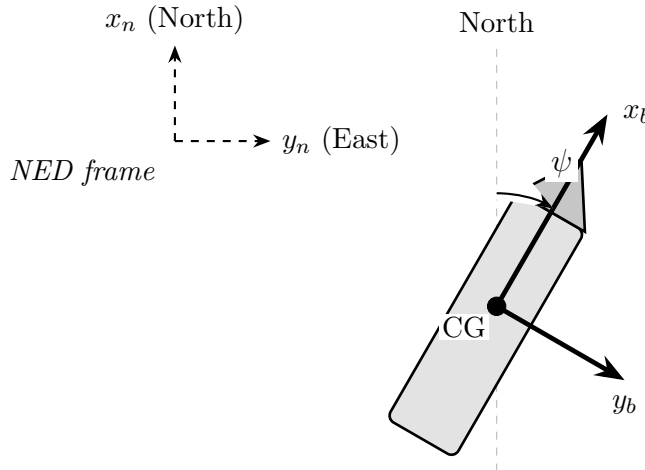


Figure 2: NED (Earth-fixed) and body (vessel-fixed) coordinate frames viewed from above. The NED frame axes x_n (North) and y_n (East) are shown as dashed arrows. The body-frame axes x_b (bow, surge) and y_b (starboard, sway) are solid. Both z axes point downward, into the page. Heading ψ is the clockwise angle from North to x_b .

Define every symbol before using it in equations or code. The report must introduce each vector and state its frame of reference.

3.2 Six-DOF Notation and Three-DOF Reduction

The standard six-DOF marine craft state vectors, in the notation of Fossen [1], are:

$$\eta_6 = [x, y, z, \phi, \theta, \psi]^\top, \quad \nu_6 = [u, v, w, p, q, r]^\top, \quad \tau_6 = [X, Y, Z, K, M, N]^\top.$$

Report Task 1. Include a notation table in your report. Explain every component and state after each vector whether it is expressed in the NED frame or the body frame.

This project uses a horizontal-plane three-DOF model. Heave, roll, and pitch are neglected; only surge, sway, and yaw are retained:

$$\eta = [N, E, \psi]^\top, \quad \nu = [u, v, r]^\top, \quad \tau = [X, Y, N_z]^\top.$$

Report Task 2. Write the three reduced vectors explicitly and state which frame each is expressed in. This frame distinction is essential: vessel dynamics compute accelerations in body axes, while position is updated in NED.

3.3 Kinematic Equation

The kinematic equation converts body-frame velocity to NED position and heading rates. It is a geometric transformation only, not a force model. For three-DOF it reduces to a yaw-only rotation:

$$\dot{\eta} = J(\psi)\nu, \quad J(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Scalar form: $\dot{N} = u \cos \psi - v \sin \psi$, $\dot{E} = u \sin \psi + v \cos \psi$, $\dot{\psi} = r$.

The inverse $J^\top(\psi) = J^{-1}(\psi)$ converts NED vectors to the body frame (e.g. a NED position error becomes a body-frame force direction).

Heading errors must always be wrapped to $(-\pi, \pi]$:

$$\text{wrap}(\alpha) = \text{atan2}(\sin \alpha, \cos \alpha).$$

Report Task 3. Write the full 6×6 Jacobian $J(\eta_6)$, then extract the 3×3 case above. Write the three scalar equations. Describe where $J^\top(\psi)$ is applied in your implementation.

The key structural point: the plant model gives $\dot{\nu}$ in body coordinates, while the kinematic equation gives $\dot{\eta}$ in NED coordinates. These two together form the state derivative needed for integration.

3.4 Process Plant Model (PPM)

Process Plant Model (PPM), in literature also known as simply Plant Model (PM) [2], is a detailed simulation model of the real physical process. Its purpose is not primarily control design, but to reproduce the real system closely enough to test whether the control system works under realistic conditions. The general six-DOF marine craft dynamics are:

$$M\dot{\nu}_6 + C(\nu_6)\nu_6 + D(\nu_r)\nu_r + g(\eta_6) = \tau_6.$$

Report Task 4. Define all terms in the six-DOF plant model and briefly state their physical meaning.

Current is handled through relative velocity. The NED current vector is rotated to the body frame and subtracted from the vessel velocity:

$$\nu_c^n = [V_{c,N}, V_{c,E}, 0]^\top, \quad \nu_c^b = J^\top(\psi) \nu_c^n, \quad \nu_r = \nu - \nu_c^b.$$

Report Task 5. Define all current variables and their coordinate frames. State whether the current direction is defined as “from” or “towards”.

For numerical simulation, the plant equation must be rearranged to solve for the vessel acceleration. The simulator state is $x = [\eta^\top, \nu^\top]^\top$.

Report Task 6. Rearrange the plant equation for $\dot{\nu}$ and write the complete state derivative $\dot{x}$:

$$\dot{\nu} = M^{-1}(\dots), \quad \dot{x} = \begin{bmatrix} J(\psi)\nu \\ M^{-1}(\dots) \end{bmatrix}.$$

The following implementation areas are especially relevant when working through Tasks 4–6: the vessel model including relative velocity, damping, Coriolis terms, and integration; the simulator loop including the ordering of reference, control, allocation, actuator limits, environmental loads, and vessel update; and the utility functions for rotations and angle wrapping.

The supplied vessel class integrates the state with the forward Euler scheme at a fixed step h ,

$$x_{k+1} = x_k + h f(t_k, x_k);$$

no other integration method is available in the Part 2 simulator, and the Part 2 engine accepts only $h = 0.1$ s or $h = 0.01$ s (the Part 1 default of 0.05 s is rejected). State the time step used in the simulation setup section of the report and keep it identical within every paired comparison.

3.5 Process Plant Model: R/V Gunnerus

The process plant used in the project is realised as a three-DOF manoeuvring model of NTNU's research vessel R/V Gunnerus, provided by `mcsimpy`. It is a zero-frequency model with constant hydrodynamic coefficients (rigid-body mass approximately 574 t), implementing

$$\dot{\eta} = J(\psi)\nu, \quad (M_{RB} + M_A)\dot{\nu} + C_{RB}(\nu)\nu + C_A(\nu_r)\nu_r + D(\nu_r)\nu_r = \tau_{thr} + \tau_{wind} + \tau_{wave} + M_A\dot{\nu}_c^b,$$

with nonlinear damping

$$D(\nu_r) = D_L + D_u|u_r| + D_v|v_r| + D_r|r|.$$

Current acts only through the relative velocity ν_r and the term $M_A\dot{\nu}_c^b$; it is not applied as a separate force. There is no restoring term $g(\eta)$ in the horizontal plane. The numerical coefficient matrices are loaded from the supplied Gunnerus data file.

Report Task 7. Relate this model to the general three-DOF equation from Task 4: explain what each term represents physically, which damping contributions are linear and which velocity-dependent, and which of these effects are dropped in the control plant model of Section 3.7.

The numbered Report Tasks end here: they cover the notation and modelling section only. Every later section states its own reporting requirements in the text, and all of them count.

3.6 Thruster Dynamics

In Part 1 the actuators were ideal: allocated thrust and azimuth commands were applied instantaneously and without limits. In Part 2 the maximum thrust, thrust-rate, and azimuth-rate limits are enabled. Students must distinguish the allocator command from the actuator state and the wrench actually delivered to the vessel.

The supplied model in part 2 is a simplified actuator model expressed directly in thrust and azimuth. The allocator requests signed thrust $u_{i,\text{cmd}}$ and, for an azimuth unit, direction $\alpha_{i,\text{cmd}}$. The actual states u_i and α_i approach these commands subject to the limits in Table 1.

For one simulation step Δt , the thrust update is

$$\Delta u_i[k] = \text{clip}(u_{i,\text{cmd}}[k] - u_i[k], -\dot{u}_{i,\text{max}}\Delta t, \dot{u}_{i,\text{max}}\Delta t), \quad (1)$$

$$u_i[k+1] = \text{sat}_{[-u_{i,\text{max}}, u_{i,\text{max}}]}(u_i[k] + \Delta u_i[k]). \quad (2)$$

Thus, a command inside the thrust bound may still be temporarily unavailable because of the ramp-rate limit. A command outside the bound produces a persistent steady-state shortfall.

For an azimuth thruster, the shortest wrapped angular error is rate limited:

$$\Delta \alpha_i[k] = \text{clip}(\text{wrap}_\pi(\alpha_{i,\text{cmd}}[k] - \alpha_i[k]), -\dot{\alpha}_{i,\text{max}}\Delta t, \dot{\alpha}_{i,\text{max}}\Delta t), \quad (3)$$

$$\alpha_i[k+1] = \text{wrap}_\pi(\alpha_i[k] + \Delta \alpha_i[k]). \quad (4)$$

The tunnel-thruster angle remains fixed. The model is a slew-rate model: the state moves linearly towards the command at the maximum permitted rate. It is not a first-order motor or propeller response.

The allocator computes a steady-state command using $B(\alpha_{\text{cmd}})u_{\text{cmd}}$, but the vessel receives the wrench formed from the *actual* actuator states:

$$\tau_{\text{thr}}[k+1] = B(\alpha[k+1])u[k+1]. \quad (5)$$

Consequently, even when $B(\alpha_{\text{cmd}})u_{\text{cmd}} = \tau_c$, it does not generally follow that $\tau_{\text{thr}} = \tau_c$ during a transient.

A persistent or rapidly varying residual can create tracking error, integrator wind-up, and oscillatory commands. The controller and anti-windup design must therefore be assessed using the applied wrench, not only the desired wrench.

The report shall state the model boundary and the numerical actuator limits. Part 2, Simulation 2 shall show commanded and actual thrust, commanded and actual azimuth, desired and applied wrench, and the residual wrench. Identify which limit is active in each important interval and relate the observed response to limitations.

3.7 Control Plant Model

The Control Plant Model (CPM), also known as Control Design Model (CDM), is a simplified mathematical description containing only the main physical properties of the process or plant; it can form part of the controller and is used for analytical stability analysis [4]. Write the control plant section of the report as follows:

- Control plant model equations.
- State the simplifying assumptions.
- Give the numerical matrices or state their source.
- Explain why this simplified model is adequate for controller design despite the nonlinear effects in the simulation model.

If a state-space method (LQR, pole placement) is used, also write equations in state space form:

$$\dot{x}_c = A_c x_c + B_c \tau_c, \quad y_c = C_c x_c.$$

Define the state ordering and all matrices. If no state-space method is used, omit this form.

Note. The control plant is a design tool. A simple linear control plant is appropriate for DP even when the process plant includes nonlinear effects. The gap between process plant model and control plant model is the reason why tuning is always iterative in practice.

4 Environmental Loads

Explain how current, wind, and waves are represented in the simulation. State what is an input, what is computed, and what is neglected. Current and the basic wind model originate in Part 1; the Part 2 wind extensions and irregular waves must be added to the final report. Ice loads, mooring lines, and other external forces can be disregarded.

4.1 Current Model

The current model is unchanged from Project Part 1; reuse your implementation. In the report state the convention, the transformation to body-fixed relative velocity, how the current enters the process plant, and the parameters used in the Part 2 simulations. If you change the implementation or the parameters, identify the change and explain why.

4.2 Wind Model

The Part 1 relative-wind and aerodynamic-coefficient model is mostly retained. Part 2 extends its wind input from a mean and slowly varying speed to

$$U_w(t) = \bar{U}_w + U_{w,\text{slow}}(t) + U_{w,\text{gust}}(t), \quad \beta_w(t) = \text{wrap}_\pi(\bar{\beta}_w(t) + \delta\beta_w(t)), \quad (6)$$

where $U_{w,\text{gust}}$ is a zero-mean spectral gust process and $\delta\beta_w$ is a bounded, slowly varying direction perturbation. Its excursion about the prescribed mean direction shall not exceed $\pm 5^\circ$.

The gust extension matters because a DP controller that reacts to every fast wind fluctuation can create unnecessary thrust variation, while the slowly varying direction changes the surge, sway, and yaw components of the mean load. Select a recognised gust spectrum. In the report state the spectrum, its discretisation, the random seed, and the time constants and variance of the slow processes; the values supplied with the course code are reference values you may keep, but state them. Show the realised time histories and discuss how they affect control effort and station keeping. If you retune the slow-speed parameters, give both sets of values and the reason.

4.3 Wave Model

An irregular-wave load model is supplied for Project Part 2. It uses a wave spectrum together with the supplied Gunnerus response data to generate the wave-force and moment time series. Students shall configure the significant wave height H_s , peak period T_p , mean direction, and random seed for each case. Wave direction follows the same “from” convention used for the other environmental conditions.

For a discretised spectrum $S(\omega)$, a long-crested irregular wave can be represented by

$$\zeta(t) = \sum_{j=1}^N a_j \cos(\omega_j t + \epsilon_j), \quad a_j = \sqrt{2S(\omega_j)\Delta\omega}, \quad (7)$$

where the phases ϵ_j are seeded random variables. The spectrum defines the sea surface, but it does *not* define the force on this vessel. Vessel- and heading-dependent

force response-amplitude operators (RAOs) map each component to a first-order wave-frequency wrench,

$$\tau_w^{(1)}(t) = \Re \left\{ \sum_{j=1}^N H_F(\omega_j, \beta_{r,j}) a_j e^{i(\omega_j t + \epsilon_j)} \right\}, \quad (8)$$

and quadratic transfer functions (QTFs) provide the slowly varying or mean second-order drift load. In a diagonal approximation,

$$\bar{\tau}_w^{(2)} \approx \sum_{j=1}^N Q(\omega_j, \omega_j, \beta_{r,j}) a_j^2. \quad (9)$$

Here $\beta_{r,j}$ is the wave direction relative to the vessel heading. This is why the supplied Gunnerus hydrodynamic data are needed: the same sea spectrum produces different forces on different vessels and at different relative headings.

Students shall not reimplement the supplied wave-load generator. They should configure H_s , T_p , mean wave direction, frequency components/range, and random seed; verify that the generated loads are finite and directionally reasonable; and explain how the resulting body-fixed wrench enters the process-plant equations.

The report shall distinguish between the first-order wave-frequency load, whose motion should be attenuated by the observer, and the slowly varying drift load, which the DP controller must reject.

The vessel model includes only surge, sway, and yaw, although the wave model may output a six-component wrench for logging and inspection. Students should understand the model inputs, outputs, assumptions, and limitations sufficiently to configure simulations and discuss the results.

Use only the approved Gunnerus wave-load database provided in the course repository. It is a repaired ShipX/Veres dataset, with corrected force-RAO phases and wave-drift yaw data. The accompanying verification notebook documents these corrections. Do not use the broken upstream dataset or modify the data manually.

5 DP System Architecture

Extend the Part 1 block diagram into a diagram of the complete Part 2 DP system, showing at least these new blocks:

- the irregular-wave wrench acting on the process plant;
- the observer producing LF pose, body-velocity, and bias estimates; and
- the measurement unit, which separates the true vessel state from the measured pose available to the observer and controller.

Show the coordinate frame of every signal and the execution order. In particular, with raw feedback the controller uses the measured pose and the simulated velocity; with observer feedback it uses $\hat{\eta}$ and $\hat{\nu}$ only. The observer receives pose and the specified wrench estimate, never measured velocity. Explain how this extended path creates two new feedback effects: observer filtering and lag change the state seen by the controller, while actuator limits make the force applied to the plant differ from the force requested by the controller.

6 Controller Design

Your Part 1 control law may carry over unchanged, but the conditions it works under do not. It is now in some simulations driven by observer estimates instead of the simulated state, and inherits the filtering and the phase lag of the observer. Thrust saturation and rate limits mean the wrench that reaches the vessel is not the wrench the controller asked for,

$$r_\tau = \tau_c - \tau_{\text{thr}}, \quad (10)$$

the allocation residual of Section 8.

Establish whether your controller still does its job, and retune or extend it if needed. In the report state the control law, the gains, the error frames and the anti-windup if implemented. Back every change with simulation evidence, and discuss the trade-off among tracking accuracy, overshoot, saturation and unnecessary actuator activity.

7 Reference Model

Your Part 1 reference model gains a second job in Part 2: the thrusters are now rate limited, and limiting the desired velocity and acceleration is what keeps the wrench demand within what they can follow. Establish whether your reference model does that against the actuator limits of Table 1, and retune it if needed.

In the report state the formulation, its natural frequency, damping, rate limits, initialisation and heading handling, and explain how the bandwidth is matched to the available thrust and actuator rates. If you retune, give both sets of values. Simulation 2 exercises the reference model against the constrained actuators.

8 Thrust Allocation

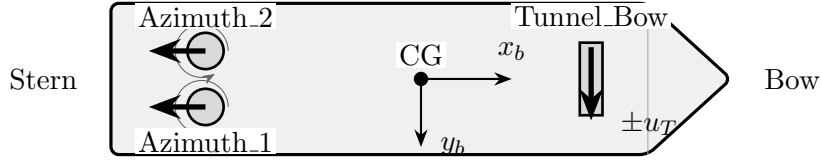
In the model for Project Part 2, constraints on rotational speed, maximum thrust and ramp-up time have been added to the thruster dynamics according to the table below. It is your job to update the thrust allocation made in Project Part 1 to take these constraints into consideration. A steady solution of $B(\alpha)u = \tau_c$ is no longer sufficient on its own: it may exceed the available thrust, and it cannot be reached immediately, because thrust and azimuth change at finite rates. Document the allocation you arrive at and relate it to the residual wrench and to the controller design, rather than presenting allocation as an isolated algebraic calculation.

8.1 Thruster Configuration

The supplied simulator uses the following three-thruster layout:

Table 1: Thruster configuration. Positions from CG in body frame ($x>0$ forward, $y>0$ starboard). Angle α_0 measured from x_b toward y_b (clockwise positive when viewed from above).

Thruster	Type	x [m]	y [m]	α_0	$u_{\max}$ [kN]	$\dot{u}_{\max}$ [kN/s]	$\dot{\alpha}_{\max}$ [°/s]
Tunnel_Bow	tunnel	+12.0	0.0	90° (fixed)	32	4	0
Azimuth_1	azimuth	-13.0	+3.0	0° (free)	80	10	12
Azimuth_2	azimuth	-13.0	-3.0	0° (free)	80	10	12



Azimuth thrusters can rotate ($\dot{\alpha}_{\max} = 12^\circ \text{ s}^{-1}$). Tunnel thruster has fixed direction.

Figure 3: Three-thruster layout (top view, body frame, bow to the right).

8.2 Part 2 Constrained-Allocation Requirements

The chosen allocation method, for example a weighted least-squares formulation [6], shall account for the limits in Table 1. The allocator receives the desired body-fixed wrench and the available actuator state and returns signed thrust and azimuth setpoints. It shall satisfy the maximum-thrust limits in its steady-state solution. When a requested wrench is infeasible, the achieved force should preserve the requested direction as far as practicable rather than being distorted by independent clipping.

The supplied actuator model enforces the transient thrust ramps and azimuth rotation limits. A rate-aware or one-step-feasible allocator is an optional extension, not a mandatory part of the basic allocation task. Clearly separate steady-state allocation from actuator dynamics in the report.

Compare desired and applied wrench, commanded and actual thrust, and commanded and actual azimuth. Define and discuss the allocation residual

$$r_\tau = \tau_c - \tau_{\text{thr}}.$$

Explain how infeasible commands and persistent residuals interact with the controller and anti-windup method if implemented.

9 Observer and Measurement Model

With the measured states from the vessel and knowing the estimated generalized forces over the vessel hull, it is possible to create an observer. The supplied measurement layer separates the true vessel state from the signals available to the observer and controller. In the mandatory simulations the pose measurement is ideal unless stated otherwise. Velocity signals may be logged for validation, but shall not be used as observer input.

The simulator has two feedback modes, selected by the observer switch of the simulation configuration. Every simulation below states which one is active.

Raw feedback (no observer). The controller receives the measured pose *and* the simulated body velocity directly (exact when sensor noise is off). This is an ideal model.

Observer feedback. The controller receives only the observer estimates $\hat{\eta}$ and $\hat{\nu}$. The observer itself receives the measured pose and the previous desired wrench, never any velocity.

The two modes therefore differ in two ways at once: the wave-frequency motion is filtered (with the associated estimate lag), and the velocity used for damping changes from the simulated value to an estimate. Keep both effects in mind when you compare them.

The observers are provided with measured position and heading only. They are *not* provided with surge, sway, or yaw-rate measurements. Each observer must generate its own velocity estimate from the pose measurements and its internal dynamic model. If pose differentiation or additional filtering is used, it is part of the observer design and must be documented as such.

Implement and tune both a nonlinear passive observer (NPO) and an extended Kalman filter (EKF); Kalman filtering has been applied to DP since the earliest systems [5], and both designs are covered in the course literature [1, 2]. Each observer shall use measured pose and an estimate of the generalized control force to provide:

- a low-frequency NED pose estimate;
- a body-fixed velocity estimate; and
- a slowly varying environmental-bias estimate if included in the chosen observer formulation.

Both observers shall separate low-frequency (LF) vessel motion from wave-frequency (WF) motion. A suitable common model is

$$\dot{\xi} = A_w \xi + E_w w_w, \quad \eta_w = C_w \xi, \quad (11)$$

$$\dot{\eta} = J(\psi) \nu, \quad M \dot{\nu} = -D \nu + J^\top(\psi) b + \tau, \quad (12)$$

$$\dot{b} = -T_b^{-1} b + w_b, \quad y = \eta + C_w \xi + v. \quad (13)$$

Here $y = [N_m, E_m, \psi_m]^\top$ is the only measured motion signal, ξ represents WF motion, b is a slowly varying NED disturbance bias, and $\nu = [u, v, r]^\top$ is an internal estimated state, not a measurement. At sample k , the observer input τ is the commanded control wrench from the previous sample, $\tau_c[k-1]$, with zero used at the first sample. This one-step delay avoids an algebraic loop and must be included when the observer equations are discretised and when convergence is interpreted.

9.1 Nonlinear Passive Observer

Derive the observer as a copy of the model above corrected by the measurement innovation, with the heading component of the innovation wrapped to $(-\pi, \pi]$. Present the resulting equations, state the numerical values of A_w, E_w, C_w, M, D, T_b and of every correction gain, and explain how the selected wave frequency and observer bandwidth separate WF motion from LF motion.

The gains of this observer are not free: the design is built on a passivity argument, and only gains that satisfy the corresponding condition inherit its stability guarantee. State the condition, show that your tuning satisfies it, and say how you established that.

9.2 Extended Kalman Filter

For the EKF, use a state containing at least WF motion, LF pose, body-fixed velocity, and slowly varying bias,

$$x = [\xi^\top, \eta^\top, \nu^\top, b^\top]^\top, \quad \dot{x} = f(x, \tau) + Gw, \quad y = h(x) + v = \eta + C_w \xi + v. \quad (14)$$

Choose a discretisation, then state and implement the predictor and the corrector you obtain from it. Give the Jacobians, the initialisation, the process covariance Q_k and the measurement covariance R_k , and write the recursion in the order your implementation actually executes it. Wrap the heading innovation before the correction.

Only R_k has a directly measurable reference. Say what each entry of Q_k represents physically, how you arrived at its value, and what the tuning costs you elsewhere in the estimate.

Simulation 4 is used to compare the two observers and select one for Simulations 5 to 7. For both designs, the report must define the model, states, input and measurement equations, discretisation, initialization, gains or covariance matrices, and tuning procedure. Validation shall examine convergence, LF tracking, WF attenuation, velocity estimation, estimate lag, and resulting actuator effort. Show explicitly how velocity is estimated from pose and distinguish the estimated velocity used in feedback from true velocity shown only as a post-run validation reference. A separate pre-filter is optional. If used, explain why it is needed.

The optional extra-credit exercise enables fixed sensor noise representing GNSS position and gyrocompass heading measurements. These prescribed noise levels are not student tuning parameters.

10 Mandatory Tests and Report

Several tests are required to show that your system is working properly. They are described in the following sections. All plots must have readable axis labels with units, legends where needed, and every figure must be discussed in the text.

For paired comparisons, use identical initial conditions, setpoints, environmental realisations, controller parameters, and evaluation intervals. Change only the component being compared. Where one run of a pair is longer than the other, evaluate both over intervals of the same length and say which they are. Sensor noise is disabled in all mandatory simulations.

Simulation 1: Environmental Loads

Disable the DP controller and let the vessel drift freely for 300 s from rest under the following combined environment:

- current 0.2 m/s from East;
- mean wind 10 m/s from North, including gust speed and slowly varying direction limited to $\pm 5^\circ$;
- waves from North-East with significant wave height $H_s = 1.5$ m and peak period $T_p = 8$ s.

This is the *design sea* of Part 2, used again in Simulations 3, 4 and 5; Simulation 6 uses a heavier one. Gunnerus has $L_{pp} = 34$ m: relate the sea state to the vessel size when you interpret the motion.

Plot position and heading time histories and the xy trajectory. Also plot the current, wind field and loads, and wave loads. The wind plots shall separate total speed, mean-plus-slow speed, gust speed, and direction. Discuss the resulting drift using the environmental inputs and forces.

Simulation 2: DP, Allocation, and Thruster Dynamics

Disable current, wind, waves, and the observer. Enable the Part 2 thruster dynamics, including maximum thrust, thrust ramp-rate, and azimuth rotation-rate constraints. Use the Part 1 four-corner sequence, holding each commanded state for 300 s:

$$\begin{aligned}\eta_0 &= [0, 0, 0]^\top, & \eta_1 &= [50, 0, 0]^\top, & \eta_2 &= [50, -50, 0]^\top, \\ \eta_3 &= [50, -50, -\pi/4]^\top, & \eta_4 &= [0, -50, -\pi/4]^\top, & \eta_5 &= [0, 0, 0]^\top.\end{aligned}$$

Here η_0 is the initial state, not a held set-point: the vessel starts at rest in η_0 , and the supplied preset commands $\eta_1, \dots, \eta_5$ for 300 s each, i.e. five legs and 1500 s in total (the same profile as in Part 1). Enable the reference model and controller. Show position and heading against their desired values, the xy trajectory, the desired controller wrench and the wrench applied by the thruster dynamics, and the commanded and actual thrust and azimuth for each thruster. If the reference model outputs velocities, plot these against the actual velocities. Discuss tracking, settling, allocation accuracy, and any intervals in which an actuator constraint is active.

Simulation 3: DP and Environmental Forces

Repeat the four-corner trajectory of Simulation 2 in the complete Simulation 1 environment. Keep the reference model and constrained thrusters enabled. Use raw feedback (measured pose and simulated velocity, Section 9); no observer shall be used.

Show position and heading against the desired trajectory, the xy plot, environmental loads, desired and applied wrench, and thruster commands and actual states. Compare quantitatively with Simulation 2 and discuss how the environment changes tracking and actuator activity.

Simulation 4: Observer Comparison and Selection

Use the Simulation 1 environment and a fixed desired body wrench

$$\tau_c = [X, Y, N_z]^\top = [2, 2, 2]^\top 10^3 \text{ [N, N, N m]},$$

about one per cent of the installed thrust, with the controller out of the loop: this constant, exactly known wrench is the τ the observers receive.

Run both observers for at least 600 s with wave loads and repeat without. Compare measured, true and estimated pose, estimated and true velocity, and the estimation errors; quantify convergence, low-frequency accuracy, wave-frequency attenuation and estimate smoothness. True velocity is validation data only. The engine resets each observer at the true initial pose, so a convergence demonstration needs an offset you introduce yourself.

Then hold the origin for 600 s in the same environment, first with raw feedback and then with the selected observer, and compare station-keeping error and actuator activity. Raw feedback also supplies the simulated velocity (Section 9). Select one observer for Simulations 5–7 and justify the choice.

Simulation 5: Thrust-Utilisation Polar Plot

Hold $[N_{SP}, E_{SP}, \psi_{SP}] = [0, 0, 0]$ with the selected observer in co-linear wind, current and waves. Sweep the common “from” direction from 0° to 350° in 10° steps, 600 s per direction, at

$$U_w = 12 \text{ m/s}, \quad U_c = 0.2 \text{ m/s}, \quad H_s = 1.5 \text{ m}, \quad T_p = 8 \text{ s},$$

using the random seed supplied with the course code, so results stay comparable between groups.

Define the instantaneous thrust utilisation from the actual thrusts as

$$U_T(t) = 100 \frac{\sum_i |u_i(t)|}{\sum_i u_{i,\max}} [\%],$$

and average it over the second half of each run unless you justify another interval.

Produce three average-utilisation polar plots: one for all directions and one for each mask below. Report both deviations per direction and discuss the three plots against each other.

- *Watch-circle mask*, as in time-domain DP capability practice: mask a direction whose maximum *low-frequency* deviation (the supplied 30 s moving-average definition) exceeds 5 m or 5° over the evaluation window.
- *Operability mask*, what a connected gangway experiences: mask a direction whose maximum *total* deviation exceeds 3 m or 3° over the same window. Horizontal motion only; a real gangway assessment also involves the vertical motions.

This is a utilisation plot for one prescribed condition, not a maximum-weather capability boundary.

Simulation 6: Observer Robustness

Use the selected observer for station keeping at the origin over 1000 s. Retain the current and wind conditions and directions of Simulation 1, and use waves from north-east with

$$H_s = 2.5 \text{ m}, \quad T_p = 10 \text{ s},$$

a heavier sea than the observer was tuned for. Keep the observer exactly as selected in Simulation 4 and reuse the same station-keeping and estimation metrics. Because this run is longer than the design-sea runs, evaluate it over an interval of the same length as the one you used there, and state which interval that is: otherwise the sea state, the run duration and the evaluation window all change together and the comparison attributes nothing. If you also want the full stress run, report it separately. Plot vessel position and heading and whatever observer quantities your robustness argument needs. Distinguish limits that come from the observer from limits that come from the actuators.

Simulation 7: DP System Functionality

Design one additional simulation that demonstrates a meaningful strength, trade-off, or limitation of the complete DP system. State the hypothesis, define all parameters, select suitable quantitative measures and plots, and discuss whether the result supports the hypothesis.

Optional Extra Credit (+3 Points): Sensor Noise

Repeat the observer comparison of Simulation 4 with the prescribed sensor noise enabled, including waves. Then repeat the closed-loop raw-versus-selected observer comparison with the same noise enabled. The fixed one-standard-deviation course values are 0.20 m for North/East position and 0.20° for heading; the simulated velocity that raw feedback uses carries 0.02 m/s and 0.05° s⁻¹; the observers never receive it. These levels are prescribed, not tuning parameters.

Discuss how sensor noise changes wave filtering, estimate noise, lag, station-keeping performance, and actuator activity. State whether the selected observer remains the preferred choice. This exercise is separate from the seven mandatory simulations.

Report Requirements

The Part 2 report shall be a *self-contained description of the complete DP system as it stands at the end of the project*. A reader should be able to understand the modelling, implementation, parameters, and results without having read the Part 1 report.

Material from Part 1 that is still relevant may therefore be reused directly in the Part 2 report, including text, equations, tables, and figures. You may shorten or revise it where appropriate. However, referring the reader to the Part 1 report instead of including the necessary information is not sufficient.

The main emphasis of the report should be on the Part 2 extensions, their physical motivation, their integration with the Part 1 system, and their effect on the closed-loop behaviour. The following sections shall be present:

- Introduction
- Coordinate frames
- Process plant model (including kinematic equation)
- Control plant model
- Current model
- Wind model (with extensions)
- Wave model
- Extended DP system architecture (original updated figure required)
- Controller and reference-model with changes or evidence that they were retained
- Observer designs (NPO & EKF), tuning, comparison, and selection
- Constrained thrust allocation
- Results and discussion for Simulations 1–7
- Optional sensor-noise results, if attempted
- Conclusion (Discussion)
- Individual contributions (factual and specific)
- Appendix: how to reproduce the Part 2 simulations

For every simulation, state the objective, setup, required evidence, and engineering conclusion. The appendix must give sufficient instructions to reproduce the submitted results.

Submit the report together with the simulation files or a repository link.

Tips

- **Work block by block.** Do not try to solve the complete system at once. Test current, wind, controller, reference model, and thrust allocation separately before running the full closed-loop simulation.
- **Perform simple sign tests.** Check that position errors, environmental forces, and body-frame transformations produce forces in the expected directions for different vessel headings.
- **Check heading wrapping.** Test setpoints near π and $-\pi$. The controller should follow the shortest rotation and should not produce an unnecessary 2π turn.
- **Use consistent units.** Use metres, seconds, radians, newtons, and newton-metres throughout the implementation. In particular, avoid mixing degrees and radians.
- **Check the thrust-allocation matrix.** Verify that the extended configuration matrix B_e , the one that maps your allocation variables to the three-DOF wrench, has $\text{rank}(B_e) = 3$ before running long simulations, and check for static thrust-limit violations or infeasible force demands.
- **Build the constrained allocation in stages.** Do not attack it all at once. Start with the azimuth angles held at fixed, sensible directions: B_e is then constant, and mapping the desired wrench to thrusts is a single linear least-squares solve. On that fixed geometry, confirm that the achieved wrench reproduces the demand through ideal actuators before you add anything else. Then let the angles move; treating each azimuth as its two body-frame force components keeps the mapping linear, and the thrust magnitude and direction follow from those components afterwards. Only then bring in the limits of Table 1: check whether the steady-state solution respects every unit's maximum thrust, and decide what your allocator does when it does not. State the rule you chose and what it preserves and sacrifices.
- **Keep the transient out of the allocator.** The supplied actuator model applies the thrust-ramp and azimuth-rotation limits whatever you command, so a steady-state allocation is enough for the basic task. Rather than trying to prevent the lag, quantify the residual between the demanded and the applied wrench and discuss what it does to tracking and to your integrator. A rate-aware allocator is an optional extension, not a requirement.
- **Present the results clearly.** Figures and simulation results should contain enough information for the reader to understand what was simulated, under which conditions, and how the system performed. Remember that the reader only knows what is presented in the report.
- **Treat the project as an engineering task.** A good implementation is important, but so are a clear system structure, readable plots, suitable interfaces, and a well-written report.
- **Plan your group work.** Even with four group members, the project can be surprisingly time-consuming alongside other course work. Start early and avoid leaving the main integration and testing to the last weekend.

A Wind Coefficients

Wind coefficients as a function of relative wind angle α_{rw} are supplied with the course code as a tabulated data set (Fig. 4). Plot the interpolated coefficients yourself and check the interpolation quality before connecting the wind model to the DP loop.

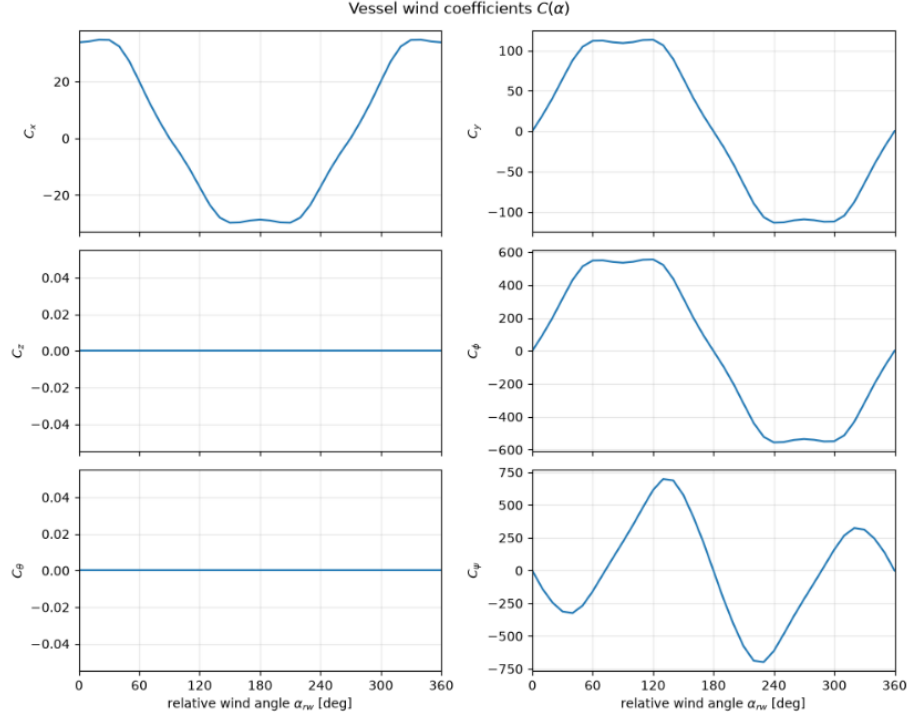


Figure 4: Wind coefficients

B Observer Design Data

The 3-DOF (surge, sway, yaw) rigid-body plus added-mass matrix and the linear damping matrix of R/V Gunnerus, as used by the simulator, are

$$M = M_{RB} + M_A = \begin{bmatrix} 6.007 \times 10^5 & 0 & 0 \\ 0 & 7.067 \times 10^5 & -4.733 \times 10^5 \\ 0 & -5.712 \times 10^5 & 5.456 \times 10^7 \end{bmatrix},$$

$$D = \begin{bmatrix} 1.118 \times 10^3 & 0 & 0 \\ 0 & 2.229 \times 10^4 & 0 \\ 0 & 0 & 1.950 \times 10^6 \end{bmatrix},$$

in SI units. They are derived from the supplied vessel database, and the course code documents that derivation with the full matrices. Note that the added-mass sway–yaw coupling is not symmetric in the vessel data; if your observer or control design requires a symmetric positive-definite mass matrix, state and justify the simplification you apply.

References

- [1] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester, UK: Wiley, 2011. [*Primary reference. Key chapters: 2 (notation), 6–7 (vessel dynamics), 8 (env. loads), 12 (DP), 13 (control design).*]
- [2] A. J. Sørensen, *Marine Cybernetics: Towards Autonomous Marine Operations and Systems*. Lecture Notes, Dept. Marine Technology, NTNU, Report UK-18-76, 2018. [*TMR4240 lecture notes. Key chapters: 6 (modelling), 7 (DP design).*]
- [3] A. J. Sørensen, “A survey of dynamic positioning control systems,” *Annual Reviews in Control*, vol. 35, no. 1, pp. 123–136, 2011. [*Good overview of DP architecture, operational modes, and control approaches.*]
- [4] A. J. Sørensen, S. I. Sagatun, and T. I. Fossen, “Design of a dynamic positioning system using model-based control,” *Control Engineering Practice*, vol. 4, no. 3, pp. 359–368, 1996. [*Foundational paper on model-based DP design; describes the control plant approach used in this project.*]
- [5] J. G. Balchen, N. A. Jenssen, and S. Sælid, “Dynamic positioning using Kalman filtering and optimal control theory,” in *Proc. IFAC Conf. Automation in Offshore Oil Field Operation*, Bergen, Norway, 1980. [*The original DP control paper. Introduced Kalman filtering and LQG design to dynamic positioning.*]
- [6] O. J. Sjørdalen, “Optimal thrust allocation for marine vessels,” *Control Engineering Practice*, vol. 5, no. 9, pp. 1223–1231, 1997. [*Describes the weighted least-squares thrust allocation approach.*]